In Search of a Convincing Root Explanation

A thinking exercise on TFR trends — Part 2

Adding layers: thinking about what lies outside the TFR data

Explanations for why TFR keeps falling

1 Lower infant mortality

No longer a need to "have many children" to secure labor or heirs

2 Women's higher education & employment

Careers and childbearing collide; later marriage and later births everywhere



3 Rising cost of raising children

A shift to "quality": heavy investment in a few children naturally limits their number

4 Urbanization & lifestyle change

Children who were "labor" on farms become "consumers" who raise living costs in cities

To reject “if it continues”, you need a mechanism that changes the direction, and evidence.

If you accept “if it continues”, go to the next step.

›

To state that mechanism, you need to know the direct causes of the **TFR** decline.

›

But the four usual explanations explained nothing.

? **Search for a more fundamental explanation.**



We will probably pick likely factors and investigate them. → Something will become clear.



Ground rules for “thinking” when investigating candidate factors

What we found

- What you find by looking it up
- What the data allow you to say

What became unclear from there

- New questions generated by what we found
- Finding facts is the start; new questions are the result

Each theme climbs a three-step ladder

Step 1

What you find by looking it up. Most people stop here.

›

Step 2

Contradictions and questions raised by Step 1. Step 1 doesn’t settle it.

›

Step 3 Proximate factors

“Who decides to have a child, and on what basis?”

01

Who decides to “have a child”?

In fact the hardest question — yet the one at the heart of the problem.

Here, we start with surface findings.

Units of decision: from the individual to the state

Individual	Couple	Family	Kin / tribe	Local / religious community	State

“The individual decides whether to have a child”
Sounds natural

“The group decides whether to grow or shrink”
Merely a metaphor; explains no mechanism

? All the layers between the two poles really exist. At which layer, and how, is the decision to have a child made?

Do vertical values (across generations) and horizontal values (within a generation) affect that decision?

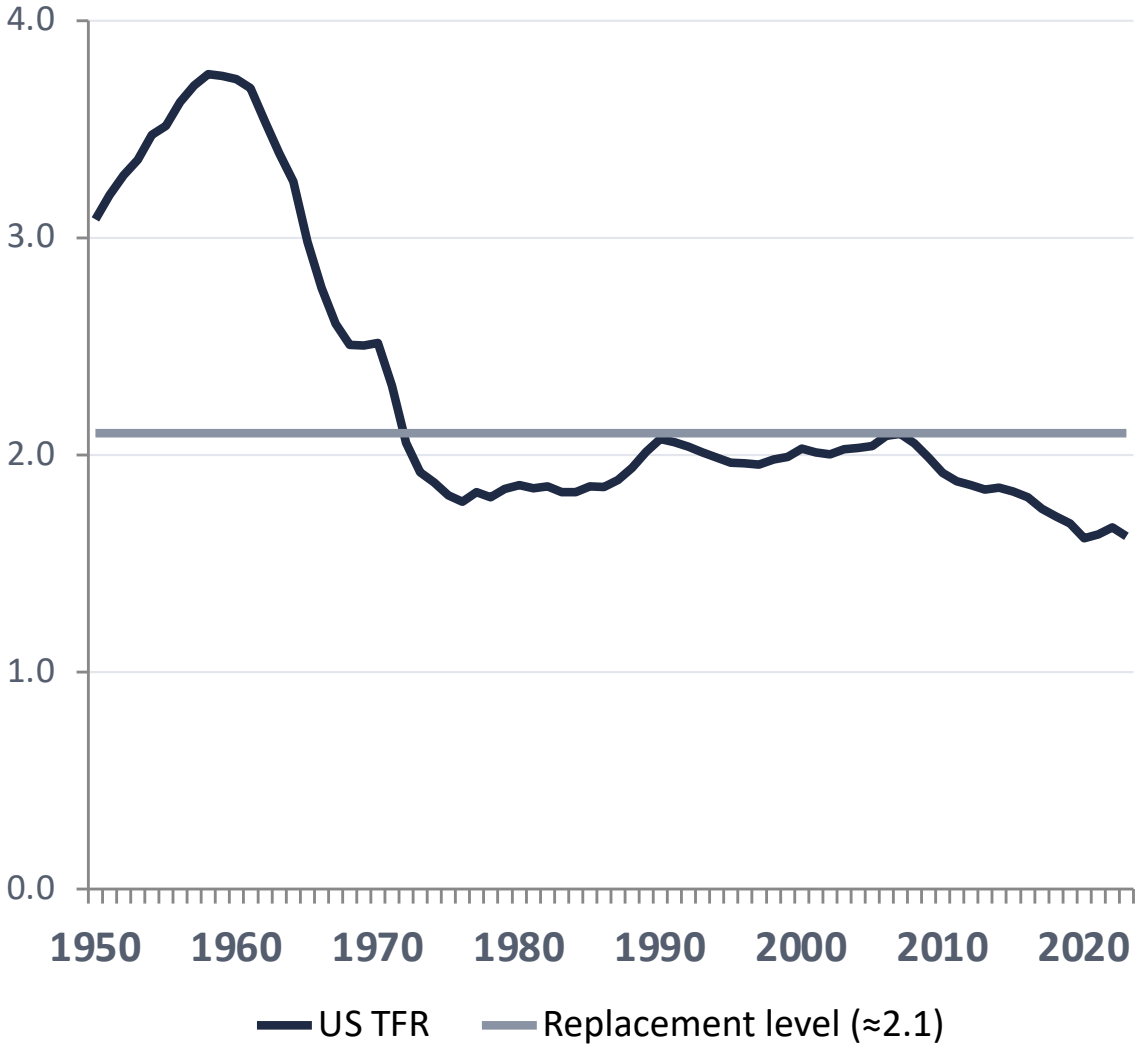
This question runs through all of Part 2. First, just feel that it cannot be answered right away.

02

Case study: the United States

Data

US TFR, 1950–2024



What we can read

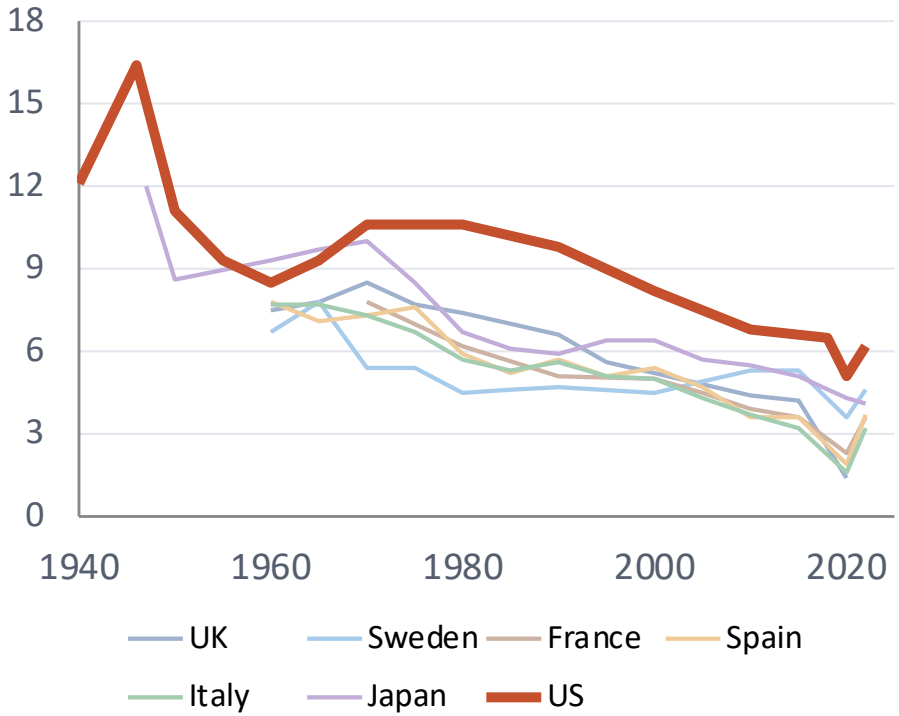
- 1950: 3.09; peak 3.75 in 1958 (baby boom)
- Sharp fall in the 1970s, to 1.79 in 1976
- Then stable around 2 for ~30 years (2.10 in 2007)
- Falling again since 2007: 1.62 in 2023

Source: UN World Population Prospects 2024 (wpp2024).

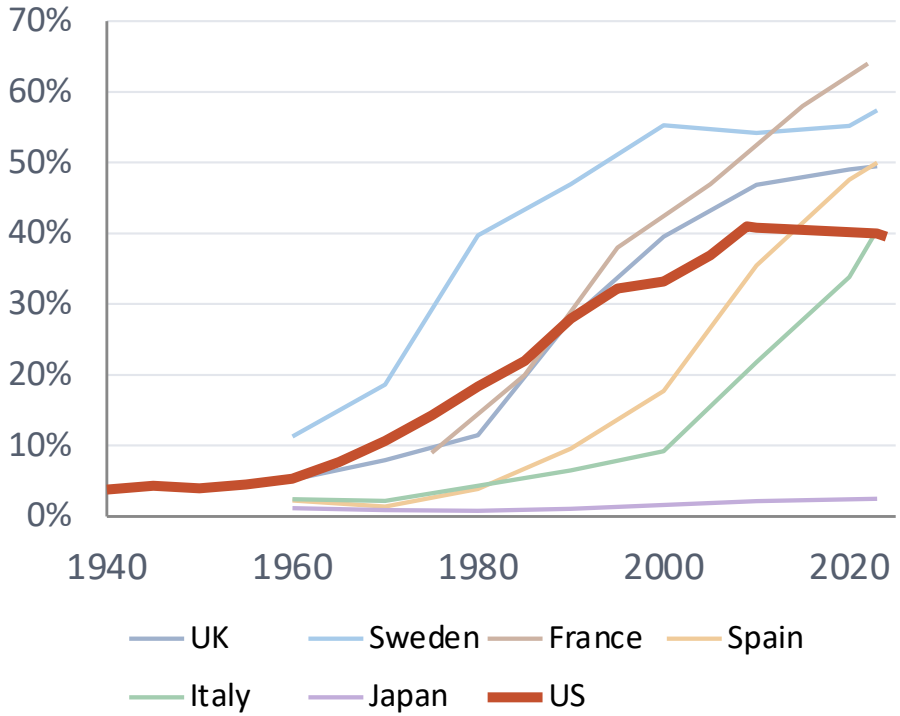
Data

The US: marriage and birth come apart

① Marriage rate (per 1,000 people)



② Share of births outside marriage

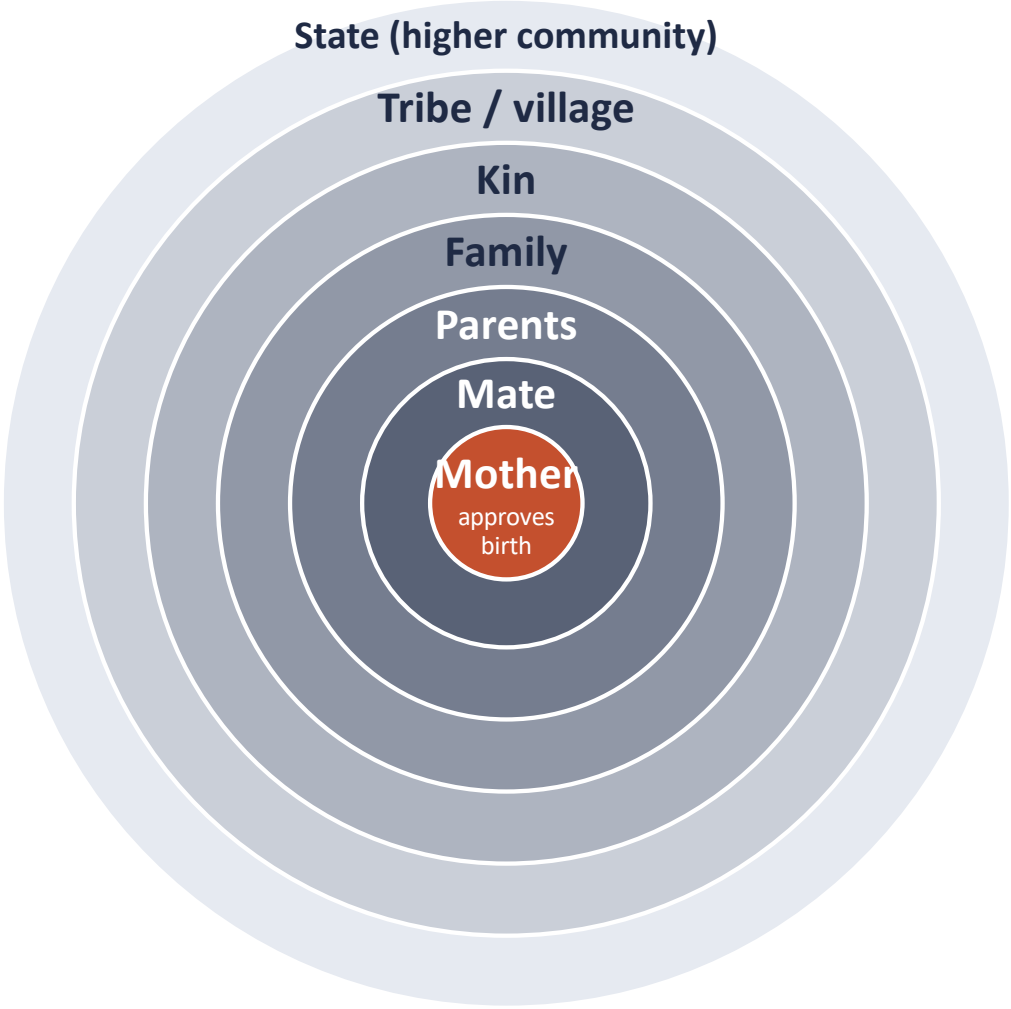


- US marriage peaked in 1946 and has long declined; Europe and Japan decline similarly from the 1970s
- US births outside marriage: 4–5% until the 1950s, then a sharp rise to a plateau of 41% in 2009. Europe has the same shape at different times; only Japan stays low

Crude marriage rate = marriages ÷ mid-year population × 1,000. US: CDC/NCHS; others: OECD Family Database (via OWID), INSEE, INED, IPSS. 2020 dip = COVID-19.

Key concept

Marriage is “approval”



Inner rings are easy to notice. Outer rings are harder to see — but they are there.

Marriage is how the group “approves” reproduction

Malinowski (1930): “principle of legitimacy”

If not approval —

why are births celebrated?

Naming ceremonies, baptisms, shrine visits…

Open Question

“Marriage” as a system of approval is fading.

Has approval itself vanished?

→ Look at the US as an example

How does marriage work on births?



Factors to investigate

Entry into marriage (ever-married share)

Share ever married by age 45 or 50 (= 1 – never-married share)

Continuity of marriage

Share of first marriages not ended by divorce

Mother's status at birth

Married / cohabiting / no partner

Number of children

Completed fertility (children born over a lifetime); **TFR**

Changing approval of births, and constraints on births

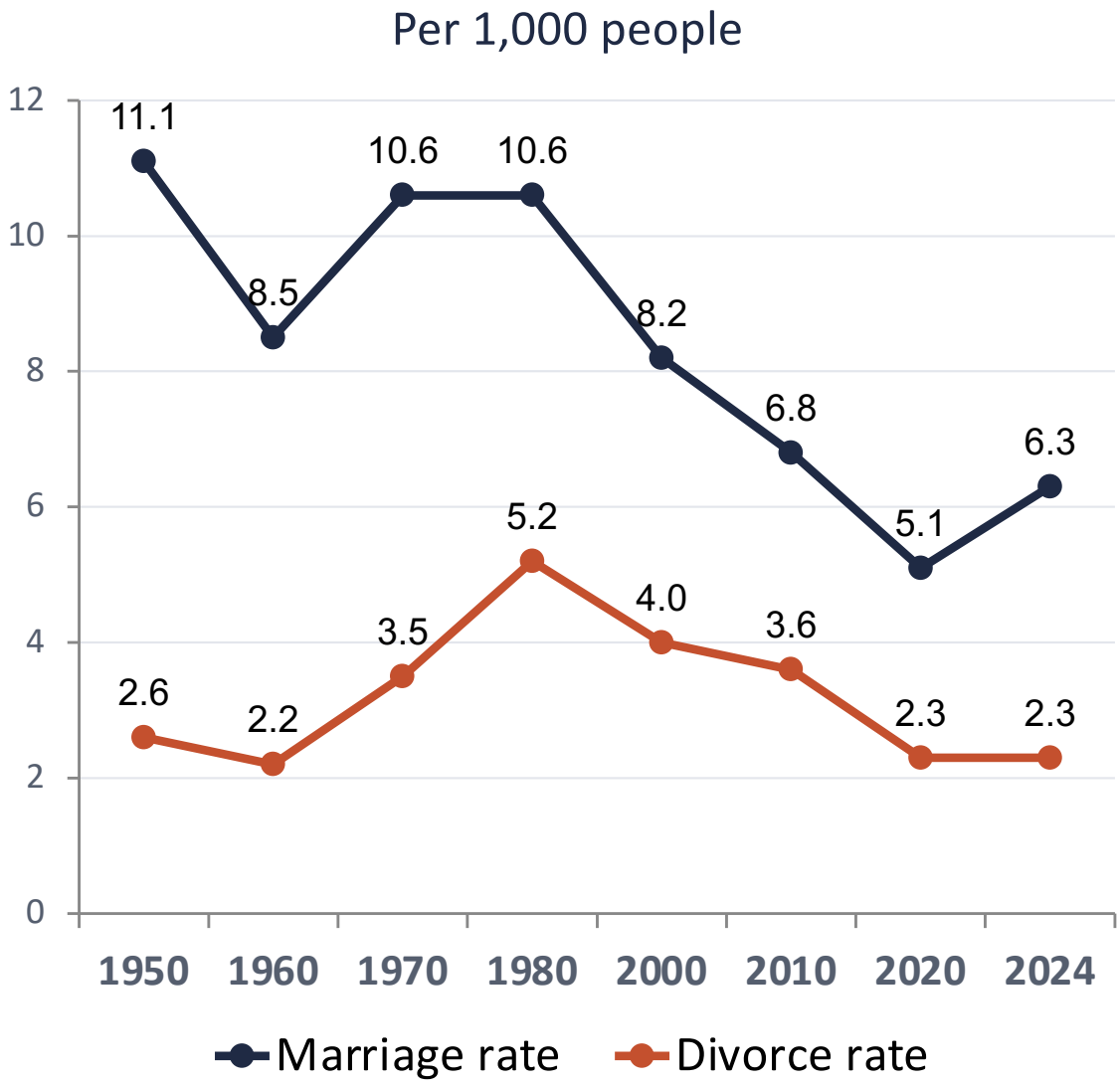
Step 1 Do groups whose marriages last have more children? Is the direction the same for men and women, and across time?…

Step 2 If some groups tie marriage to childbearing and others separate them, who decides to “have a child” in each?

Step 3 By education / income (wealth): how do marriage entry, continuity, mother’s status at birth, and number of children differ?

Data

US marriage and divorce, 1950 and 2025



Median age at first marriage	1956	2025
Women	20.1	28.4
Men	22.5	30.8

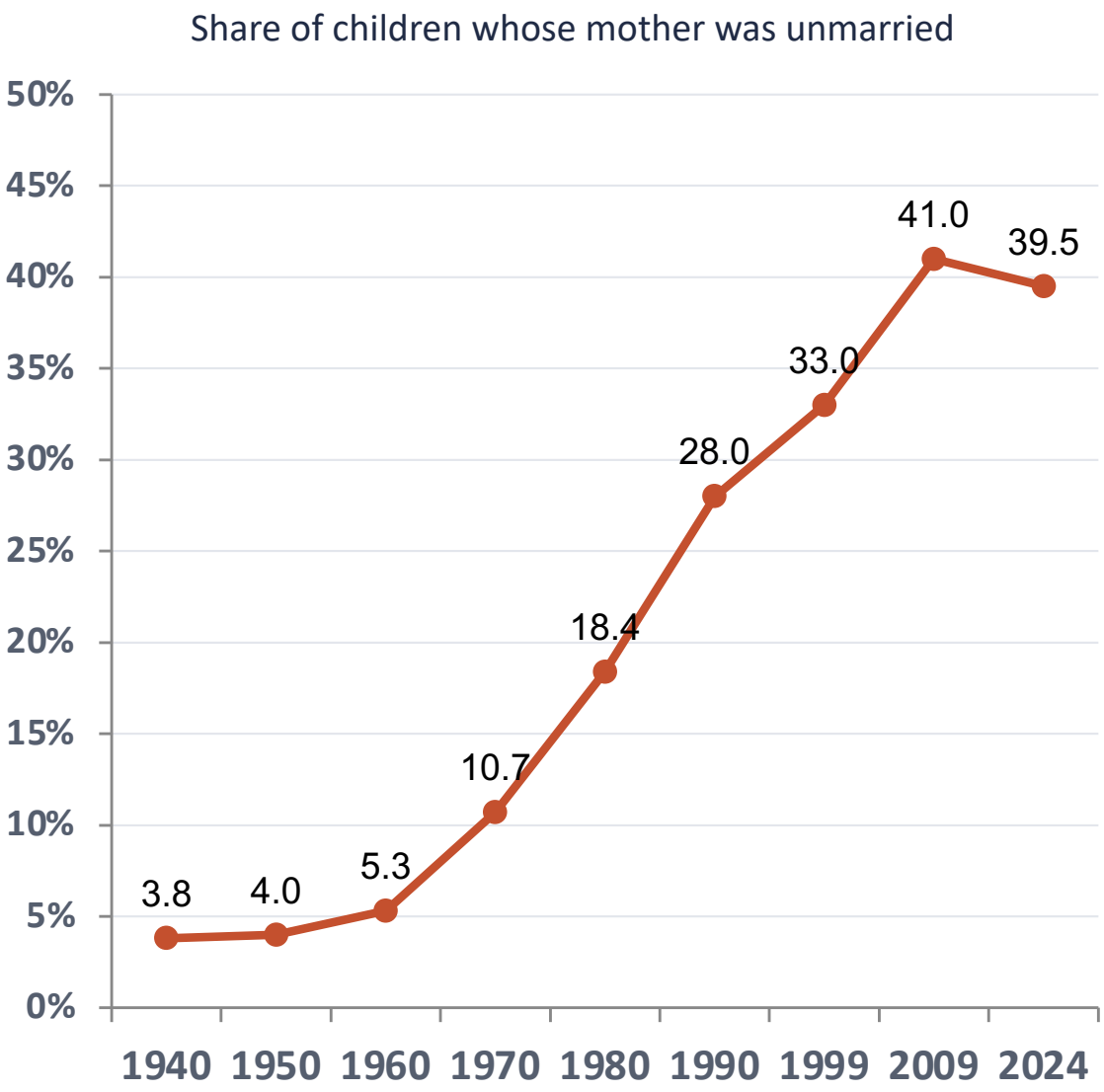
What we can read

- Marriage came later, and less often
- Divorce doubled by 1980 vs 1950, then returned to its old level
- As marriage became rarer, it also became more stable

Source: CDC/NCHS (marriage and divorce rates; 2024 provisional; divorce excludes some states); NCFMR FP-26-13 (age at first marriage).

Data

Births to unmarried mothers (US, since 1940)



Race	1965	1990	2024
White	3.1%	18%	26.2%
Black	24%	64%	68.6%

What we can read

- In 1950, children were born almost entirely within marriage (4% unmarried)
- A sharp rise from the 1970s, peaking at 41% in 2009
- Rose among both whites and blacks; the gap was there from the start

Source: NCHS NVSR 48-16 (1940–99), NCHS Data Brief 162 (2009), Births: Final Data for 2024, Brookings (by race 1965, 1990; 2024 non-Hispanic).

Data

Births and marriage by race and ethnicity (US, 2024)

Group	TFR	Births to unmarried mothers	Unmarried women's birth rate (‰)
All	1.60	39.5%	35.5
White (non-Hisp.)	**1.53**	**26.2%**	**24.1**
Black (non-Hisp.)	**1.50**	**68.6%**	**46.4**
Hispanic	1.93	54.2%	56.4
Asian (non-Hisp.)	1.26	11.5%	11.4
American Indian / Alaska Native	1.38	68.6%	47.0
Native Hawaiian / Pacific Islander	2.19	53.7%	71.5

What we can read

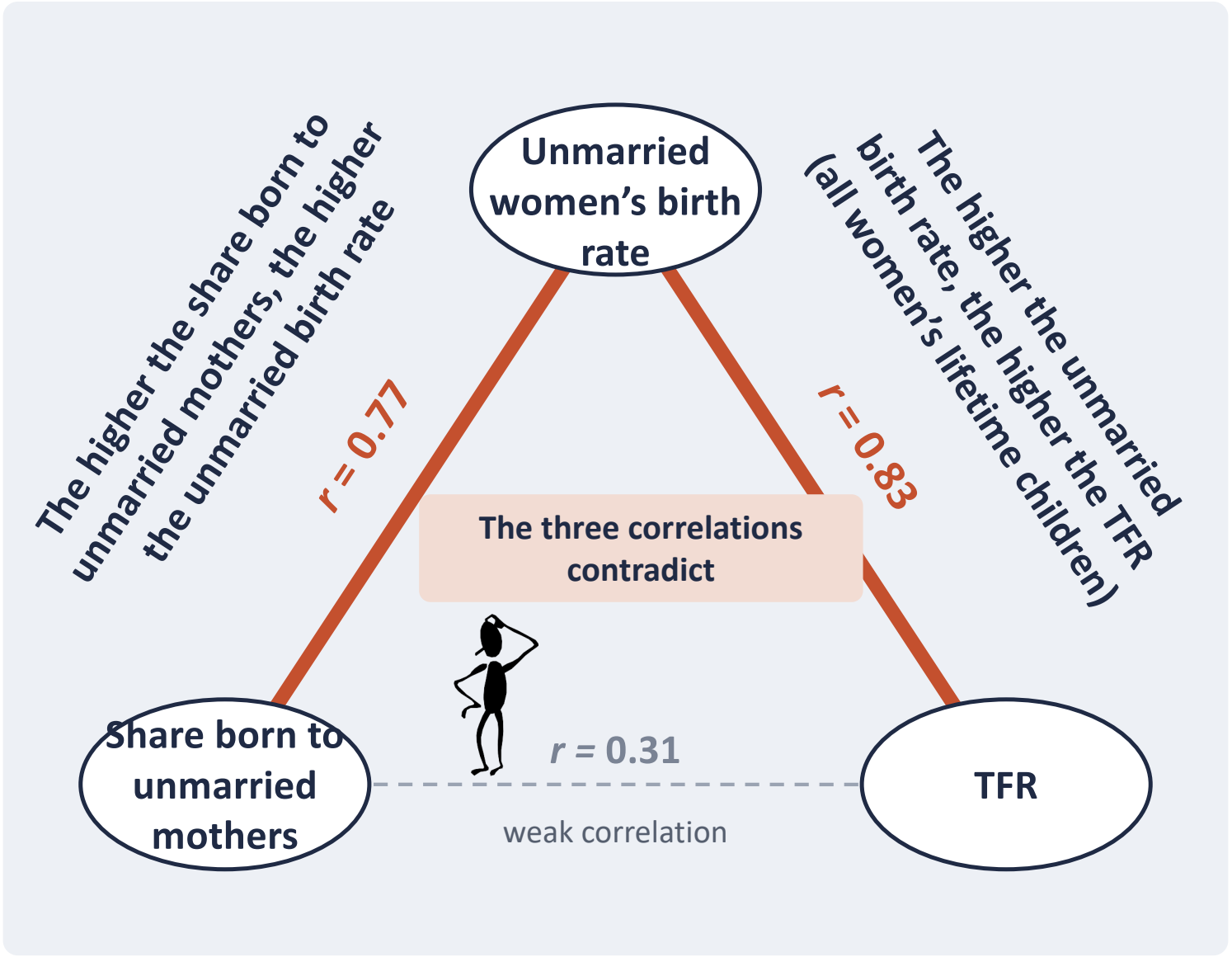
- White vs Black: births to unmarried mothers 26% vs 69% — very different
- Yet TFR is 1.53 vs 1.50 — almost the same

?

How are race group, family form and TFR related?



Source: CDC/NCHS, Births: Final Data for 2024 (NVSR 75-2). Unmarried women's birth rate = births per 1,000 unmarried women aged 15–44.

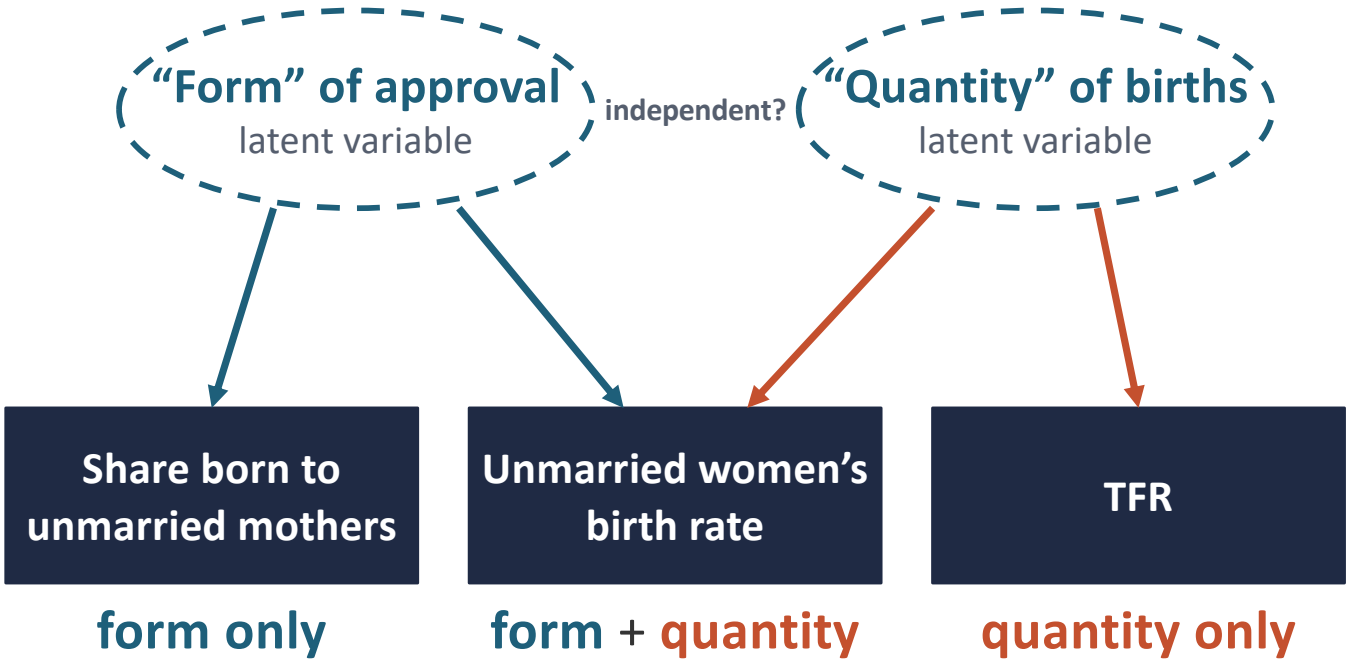




Something unseen must be in between

Think

Inference: two unseen variables sit in between



"Form" of approval

Is it normal to pass through marriage when giving birth, or to give birth with only the approval of a partner, family or community?

"Quantity" of births

How many children a group's women have over a lifetime — set by economics, desired family size, etc.; works inside and outside marriage

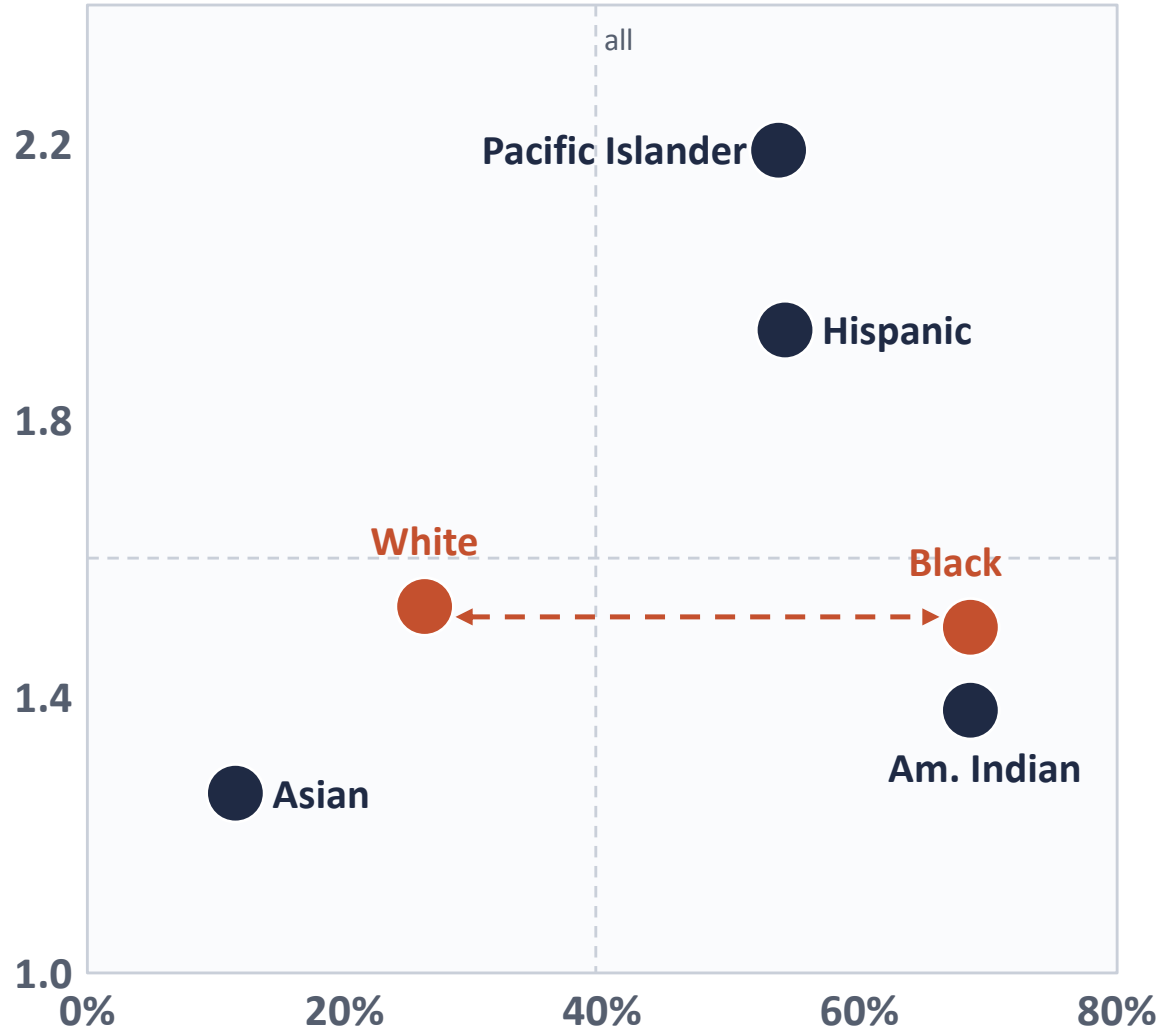
This resolves the contradiction

- Unmarried women have many children only when the group has many (quantity) AND births outside marriage are normal (form) — so this rate correlates with both
- The share reflects only form; TFR only quantity. If form and quantity are independent, they do not correlate
- From 0.77 and 0.83, math says only "the third is ≥ 0.28". The observed 0.31 is near that floor

Think

Map of quantity and form: where is each group?

TFR = "quantity"



Births to unmarried mothers = "form" (right = outside marriage)

Why is each race group's point where it is?

Infer the "why" by activating knowledge

An adult with some knowledge will think, "Ah, maybe…"

Sociologist's eye: immigration, inequality, institutions, labor

Anthropologist's eye: kinship, approval practices, history

Dashed lines = all-US values (share 39.5%, TFR 1.60). Source: CDC/NCHS, Births: Final Data for 2024 (NVSR 75-2).

Think

Inferring the "why" of each group's position

White
form: marriage / quantity: mid

- The majority; the baseline for comparison
- Split by education: graduates marry first; others more often outside marriage

Black
form: outside / quantity: mid

- Conditions for marriage hard to meet: male joblessness, incarceration
- History since slavery
- Grandmothers and kin networks support childrearing

Hispanic
form: outside-leaning / quantity: high

- Immigration; consensual-union (unión libre) tradition
- Familism (familismo)
- Young population, earlier births

Asian
form: marriage / quantity: low

- Many highly educated, skilled immigrants (selection)
- Fewer children, heavy investment
- East/South Asian norm: marry, then give birth

Am. Indian
form: outside / quantity: lowish

- Poverty on reservations
- History of intervention in family and culture (boarding schools)
- Tribal customary marriages may go unregistered

Pacific Islander
form: outside-leaning / quantity: highest

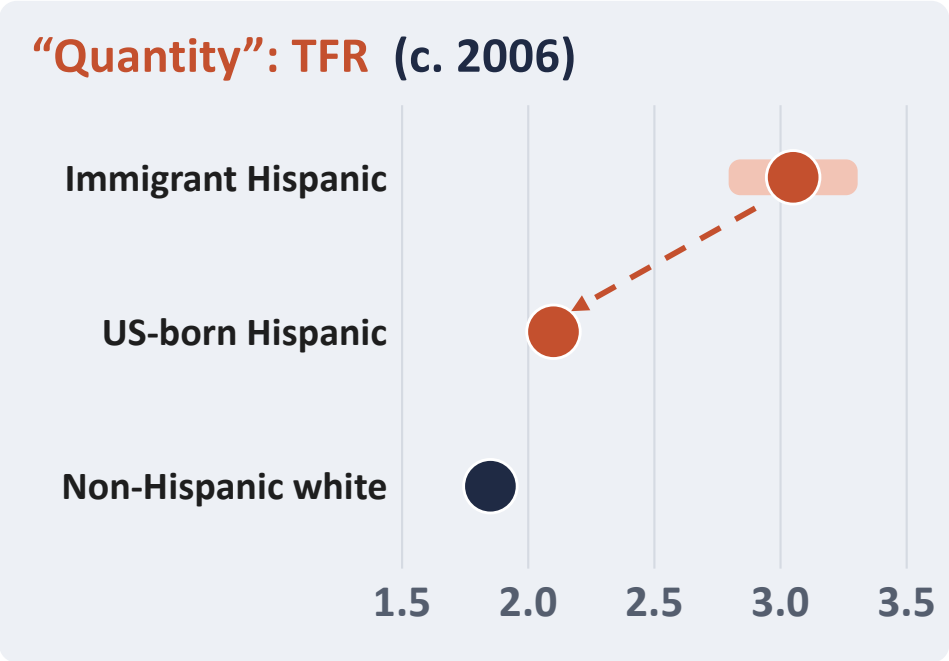
- Young population; high fertility in origin societies
- Extended-family childrearing (e.g., Hawaiian hānai)

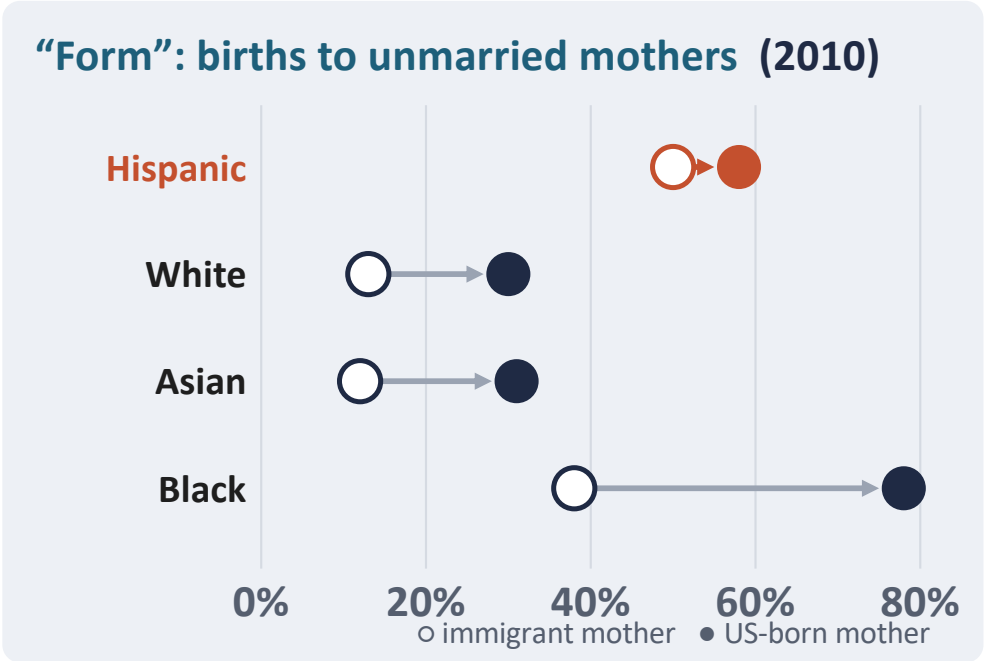
Use: describe the position → hypothesize from knowledge → if true, what else should we see? → check within-group spread.
Caution: knowledge can become prejudice.

Extension

Immigrant generations: where do the points move?

Immigrant mothers (1st generation) vs US-born mothers (2nd+ generation)





What we found

- Quantity converges: US-born Hispanics' TFR nears whites'
- Form does not: in every group, US-born mothers give birth outside marriage more often
- "Assimilation" runs in different directions for quantity and form

→ New questions

- Into which "layer" of US society do immigrants' children assimilate?
- What about being born and raised in the US moves the "form" outside marriage?

US-born = 2nd+ generation (2nd and 3rd+ not separable). Immigrant TFR is inflated by births soon after migration; completed fertility ≈ 2.3–2.4 (Parrado 2011). Sources: Parrado 2011, Demography 48(3); Pew Research Center 2012 (NCHS data).

Summary

Interim summary: back to the fall of US TFR

What we found

- The “form” of approval by marriage faded from the 1960s; births outside marriage plateaued at ~40%
- “Form” and “quantity” are separate axes
- Since 2007, form barely moved; only quantity fell, in every group
- Hispanic growth held TFR up until 2007; ¾ of the fall since came from whites and Hispanics

How we found it

- Long trends, overlaid with other countries
- From the correlation triangle, inferred two unseen variables
- Activated knowledge to hypothesize each group’s position
- Compared immigrants and US-born

Still unknown

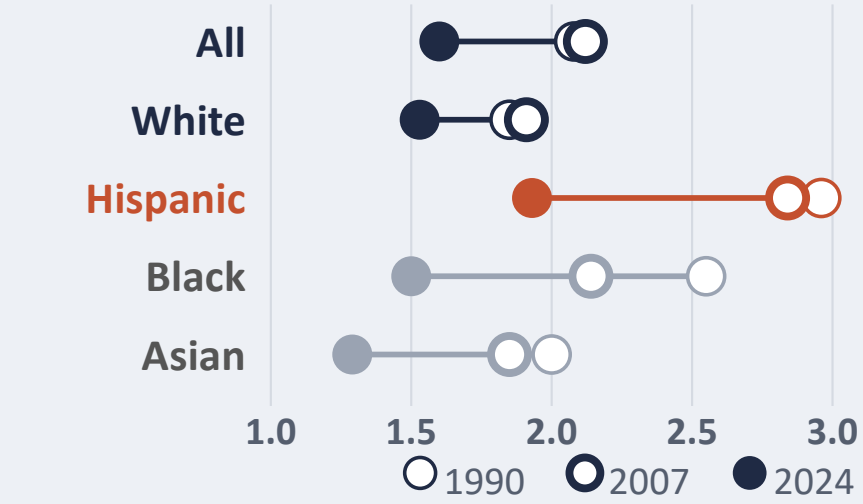
- What is lowering “quantity”?
- Is the Hispanic plunge due to less immigration and generational change?
- For whites: later marriage, more never-married, or fewer births within marriage?
- If form is unchanged but quantity falls, how are approval and quantity linked?

Data

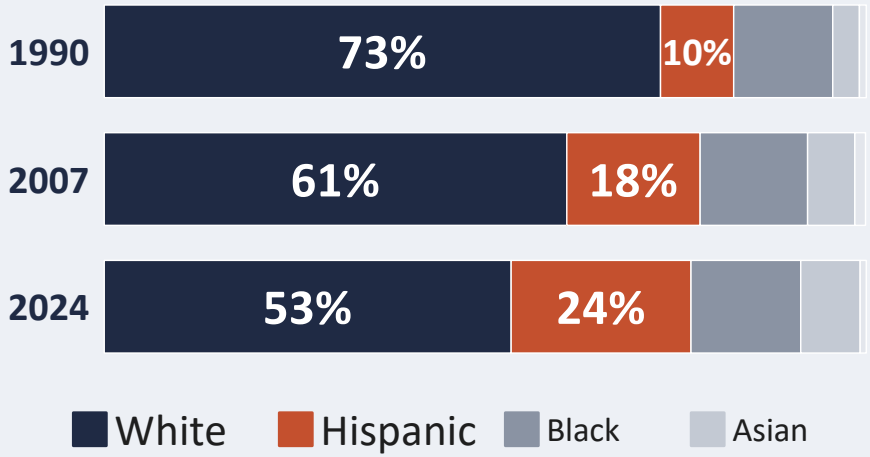
Key: whites & Hispanics — weight × quantity

Three points: 1990 (Hispanic statistics complete) · 2007 (TFR peak) · 2024 (latest)

① Quantity: TFR



② Weight: women aged 15–44



③ Decomposing the TFR change

1990→2007

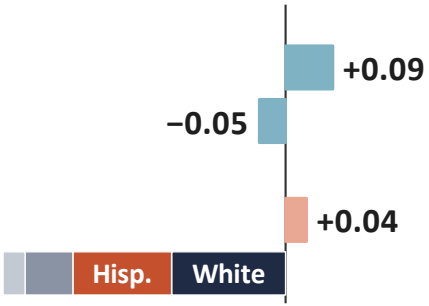
composition shift +0.09

within-group quantity −0.05

2007→2024

composition shift +0.04

within-group quantity −0.54



Until 2007, white decline and Hispanic growth cancelled out. Since then, quantity fell within groups — ¾ of it among whites and Hispanics

④ Meanwhile, "form" barely moved

Births to unmarried mothers, 2007→2024

White 27.8% → 26.2%

Hispanic 51.3% → 54.2%

What fell after 2007 was quantity itself, not the form of approval

Weights = women 15–44 derived as births ÷ general fertility rate. ③ treats TFR ≈ Σ(weight × group TFR) (approximate; off by ~0.02). "Asian" = API in 1990/2007; Asian + NHPI in 2024. Sources: Hamilton & Kirmeyer 2017 (NVSR 66-2); CDC/NCHS natality by race and Hispanic origin (data.cdc.gov); Births: Final Data for 2007, 2024.

Think

Next: decomposing white and Hispanic "quantity"

Whites (majority): large weight

Hypotheses

- Marriage and childbearing are delayed
- More people never marry
- Education split: who has fewer births within marriage?
- Housing, education costs, job insecurity

Data needed to check

- Age-specific fertility; mean age at birth
- Total first-marriage rate; ever-married by 50
- TFR by education and marital status

Hispanics: steep decline

Hypotheses

- Less Mexican immigration since 2008 → fewer immigrant women (high quantity)
- Shift to US-born generations, near white levels
- Large drop in teen / early-20s births
- Hit of the 2008 Great Recession

Data needed to check

- TFR by nativity (immigrant / US-born)
- Age-specific fertility (esp. teens)
- By origin (Mexican, Central American, Cuban…)

To the main question: the fall in US TFR is a fall in quantity, not a change in the form of approval. What sets quantity? Is it unrelated to approval — or have the "outer rings" of approval (society, the state) changed?

Big Question!

TFR keeps trending down.
Will humankind eventually disappear?

1 Lower infant mortality
No longer a need to "have many children" to secure labor or heirs

2 Women's higher education & employment
Careers and childbearing collide; later marriage and later births everywhere



3 Rising cost of raising children
A shift to "quality": heavy investment in a few children naturally limits their number

4 Urbanization & lifestyle change
Children who were "labor" on farms become "consumers" who raise living costs in cities

United States
viewed as (marriage–birth) × (race · status)
Japan
⋮

What we found
- **what we find by looking it up**
- **what the data allow us to say**



Has the Big Question been solved?
Is the explanation exhausted, the understanding sufficient?

What became unclear from there
- **Take on the new questions generated by what we found**

Looking back so far

Big Question **TFR keeps falling. Will humankind eventually disappear?**

1 The standard explanations

Lower infant mortality	Women's education & work	Cost of raising children	Urbanization & lifestyle

Plausible. But how deep do they really go…

2 Case study

US: marriage · birth × race · ethnicity

What we found

- "Form" of approval and "quantity" of births are separate axes
- Since 2007, a fall in "quantity"
- ¾ of it among whites and Hispanics

So far

Check: is the Big Question no longer a question?



How tiny!

We have not even reached “why” or “how far”

What the case study found

In which ethnic groups US TFR fell

3 **New questions born from what we found**

Within the US	To other societies	To the whole world
• What sets “quantity”? • Will the fall stop somewhere?	• Does the US pattern fit Japan? • If not, what creates the pattern?	• Separate causes, same decline? • A common root principle?

Compass: does this question bring us closer to the Big Question?

So far

Looking back: returning to the Big Question

Big Question TFR keeps falling. Will humankind eventually disappear?

1 **Standard explanations of the TFR fall** four common factors

Lower infant mortality	Women's education & work	Cost of raising children	Urbanization & lifestyle

Plausible. But whether they are true has not been checked with data.

2 **Case study** US: (marriage–birth) × (race · ethnicity)

- Form of approval faded, but form and quantity are separate axes
- The fall since 2007 is in quantity; ¾ among whites and Hispanics
- Across immigrant generations, quantity converges to the majority

✓ **Check** **Is the Big Question solved? Is the explanation exhausted?**

We found "where" it fell.
"Why" and "how far" are still far off

3 **New questions from the case study**

- What sets "quantity"? Do the outer rings of approval (society, state) matter?
- Will the fall stop somewhere? Is there a floor?
- Does the US pattern fit Japan and other societies?
- If not, is the pattern different? What generates it?
- Worldwide, do separate causes produce the same TFR decline?
- Or does a common root principle drive a common decline?

Homework

Homework: your report

Theme **Your own question within the Big Question**

"TFR keeps falling. Will humankind eventually disappear?"

What to do

1. **Choose** one or few question that brings us closer to the Big Question
2. **Investigate** what you find by looking it up, and what the data allow you to say
3. **Go further** what became unclear: the new questions your findings generate

Another way in — from the opposite side

Counting people shows where TFR fell. The why may lie in what is harder to count: literacy, religion, family systems. Emmanuel Todd reads deep social change in these. What becomes of fertility in a society that has reached his "zero religion"? Daring reasoning is welcome — if you test it against evidence.

Keep in mind

- Question, followed deep — not shallow
- Your reasoning, not a summary of facts
- A good new question beats a final answer

Use AI — and argue with it

Encouraged, but expect dozens of rounds.
It misreads, stays shallow, even fibs.
Your brain will want to ask back. **Enjoy it!**